



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
	Career Forge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline	
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Abstract

CareerForge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline is a comprehensive web-based application designed to support students in preparing for internships and employment opportunities through intelligent career guidance and modern software engineering practices. Many students face challenges in evaluating their resumes, identifying skill gaps, preparing for technical interviews, and tracking their learning progress. CareerForge addresses these challenges by providing an integrated platform that combines artificial intelligence, secure software development, cloud technologies, and machine learning operations to create a personalized career development experience.

The platform allows students to register securely, create and manage personal profiles, upload resumes, and receive AI-powered resume analysis with suggestions for improvement. Based on the extracted skills and resume content, the system recommends learning resources, certifications, and personalized career roadmaps to help students enhance their technical competencies. The platform also provides AI-assisted interview preparation by generating technical and behavioural interview questions, tracking user performance, and offering detailed feedback to improve confidence and interview readiness. A comprehensive dashboard enables users to monitor their resume score, completed learning activities, interview history, and overall career progress.

The project follows Agile Software Engineering principles throughout its development lifecycle. Requirements are captured through user stories and organized into a prioritized product backlog. Scrum methodology is adopted for sprint planning, incremental development, testing, and continuous feedback. UML diagrams, software architecture design, database modelling, and system documentation are prepared to ensure maintainability and scalability of the application. Version control is managed using Git and GitHub, enabling collaborative development and efficient code management.

To implement DevOps practices, the application is containerized using Docker and automated through GitHub Actions for continuous integration and continuous deployment (CI/CD). Cloud deployment is performed on Amazon Web Services (AWS), ensuring high availability and scalability. Application monitoring and performance analysis are achieved using Prometheus and Grafana, enabling real-time observation of system health and resource utilization.

Security is incorporated as an integral part of the software development lifecycle through DevSecOps practices. The platform implements secure authentication using Spring Security and JSON Web Tokens (JWT), password encryption using BCrypt, role-based access control, and secure API communication. Static code analysis is performed using SonarQube, while Docker container vulnerabilities are identified using Trivy. Additional security testing based on OWASP guidelines ensures protection against common web application vulnerabilities, resulting in a secure and reliable software solution.

The artificial intelligence and machine learning components are developed using Python, Scikit-learn, Pandas, and FastAPI. Machine learning models are employed to evaluate resumes, identify skill gaps, and generate personalized recommendations for career development. MLOps practices are incorporated using MLflow for experiment tracking, model versioning, deployment, and lifecycle management, ensuring that machine learning models remain reproducible, maintainable, and easy to update as new data becomes available.

Overall, CareerForge demonstrates the practical integration of Software Engineering, Agile development, DevOps, DevSecOps, Cloud Computing, Artificial Intelligence, and MLOps within a single application. The project provides students with a secure, intelligent, scalable, and cloud-ready platform that enhances career preparation while showcasing modern industry practices. It serves as a comprehensive capstone project that reflects current software development methodologies and prepares students for real-world engineering challenges.